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Пензенский государственный университет
Кафедра “Вычислительная техника”

Отчёт
по лабораторной работе №1
по курсу “Программирование на языке Java”
на тему “Графические интерфейсы”

Выполнили студент гр.

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Цель работы: научиться разрабатывать приложения, обладающие графическим интерфейсом пользователя, с использованием библиотеки Swing.

Листинг:

```
package my.l1frame;

import java.awt.Color;
import java.awt.Font;
import javax.swing.JLabel;
import javax.swing.table.DefaultTableCellRenderer;

public class l1frame extends javax.swing.JFrame {

    private static final java.util.logging.Logger logger =
        java.util.logging.Logger.getLogger(l1frame.class.getName());
    private javax.swing.table.DefaultTableModel tableModel;

    public l1frame() {
        initComponents();
        tableModel = (javax.swing.table.DefaultTableModel) jTable1.getModel();

        jButton1.setBackground(Color.WHITE);
        jButton1.setForeground(Color.BLACK);

        jButton2.setBackground(Color.WHITE.darker());
        jButton2.setForeground(Color.BLACK);

        jButton3.setBackground(Color.RED);
        jButton3.setForeground(Color.WHITE);

        jTable1.setBackground(Color.LIGHT_GRAY);
        jTable1.setForeground(Color.BLACK);

        jTable1.getTableHeader().setBackground(Color.DARK_GRAY);
        jTable1.getTableHeader().setForeground(Color.WHITE);

        jLabel1.setForeground(Color.MAGENTA);
        jLabel3.setForeground(Color.MAGENTA);
        jLabel4.setForeground(Color.MAGENTA);
        jLabel5.setForeground(Color.MAGENTA);
        jLabel6.setForeground(Color.MAGENTA);
    }

    @SuppressWarnings("unchecked")

    private void initComponents() {
```

```

jLabel1 = new javax.swing.JLabel();
jLabel2 = new javax.swing.JLabel();
jLabel3 = new javax.swing.JLabel();
jLabel4 = new javax.swing.JLabel();
jLabel5 = new javax.swing.JLabel();
jLabel6 = new javax.swing.JLabel();
jTextField1 = new javax.swing.JTextField();
jTextField2 = new javax.swing.JTextField();
jTextField3 = new javax.swing.JTextField();
jButton1 = new javax.swing.JButton();
jButton2 = new javax.swing.JButton();
jScrollPane1 = new javax.swing.JScrollPane();
jTable1 = new javax.swing.JTable();
jButton3 = new javax.swing.JButton();

setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
setAutoRequestFocus(false);
setBackground(new java.awt.Color(153, 0, 204));

jLabel1.setFont(new java.awt.Font("Sitka Small", 0, 14));
jLabel1.setText("Функция для расчета:  $e^x/x$ ");
jLabel1.setCursor(new java.awt.Cursor(java.awt.Cursor.SE_RESIZE_CURSOR));

jLabel3.setFont(new java.awt.Font("Sitka Small", 0, 12));
jLabel3.setText("");

jLabel4.setFont(new java.awt.Font("Sitka Small", 0, 12));
jLabel4.setText("Верхний предел:");

jLabel5.setFont(new java.awt.Font("Sitka Small", 0, 12));
jLabel5.setText("Нижний предел:");

jLabel6.setFont(new java.awt.Font("Sitka Small", 0, 12));
jLabel6.setText("Шаг:");

jTextField1.addActionListener(this::jTextField1ActionPerformed);

jButton1.setText("Заполнить");
jButton1.addActionListener(this::jButton1ActionPerformed);

jButton2.setText("Рассчитать");
jButton2.addActionListener(this::jButton2ActionPerformed);

jTable1.setModel(new javax.swing.table.DefaultTableModel(
    new Object [][] {
    },
    new String [] {
        "Нижний предел", "Верхний предел", "Шаг", "Результат"
    }
));

```

```
jTable1.addComponentListener(new java.awt.event.ComponentAdapter() {
    public void componentHidden(java.awt.event.ComponentEvent evt) {
        jTable1ComponentHidden(evt);
    }
});
jScrollPane1.setViewportView(jTable1);

jButton3.setText("Удалить строку");
jButton3.addActionListener(this::jButton3ActionPerformed);

javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .add(layout.createParallelGroup(
                javax.swing.GroupLayout.Alignment.LEADING
            ).add(jLabel5, javax.swing.GroupLayout.PREFERRED_SIZE, 117,
                javax.swing.GroupLayout.PREFERRED_SIZE)
                .add(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE, 71,
                    javax.swing.GroupLayout.PREFERRED_SIZE))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .add(jLabel2, javax.swing.GroupLayout.PREFERRED_SIZE, 37,
                javax.swing.GroupLayout.PREFERRED_SIZE)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .add(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE, 218,
                javax.swing.GroupLayout.PREFERRED_SIZE)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .add(jLabel6, javax.swing.GroupLayout.PREFERRED_SIZE, 117,
                javax.swing.GroupLayout.PREFERRED_SIZE)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .add(jTextField3, javax.swing.GroupLayout.PREFERRED_SIZE, 71,
                javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGroup(layout.createParallelGroup(
                javax.swing.GroupLayout.Alignment.TRAILING
            ).add(jLabel4, javax.swing.GroupLayout.PREFERRED_SIZE, 117,
                javax.swing.GroupLayout.PREFERRED_SIZE)
                .add(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE, 71,
                    javax.swing.GroupLayout.PREFERRED_SIZE)
                .add(jLabel3))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .add(jButton1)
        )
);
```

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        .addComponent(jButton2))
        .addGap(57, 57, 57))
        .addComponent(jScrollPane1, javax.swing.GroupLayout.DEFAULT_SIZE, 554,
Short.MAX_VALUE)
        .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
layout.createSequentialGroup())
        .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
        .addComponent(jButton3)
        .addGap(30, 30, 30))
    );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup())
        .addComponent(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE, 48,
javax.swing.GroupLayout.PREFERRED_SIZE)
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
        .addComponent(jLabel3)
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
        .addComponent(jLabel2)
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)

.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jLabel4)
        .addComponent(jTextField1, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(18, 18, 18)

.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jLabel5)
        .addComponent(jTextField2, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
        .addComponent(jButton1))
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)

.addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(jLabel6)
        .addComponent(jTextField3, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, javax.
swing.GroupLayout.PREFERRED_SIZE)
        .addComponent(jButton2))
        .addGap(35, 35, 35)
        .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 183,
javax.swing.GroupLayout.PREFERRED_SIZE)
        .addGap(18, 18, 18)
        .addComponent(jButton3)
        .addContainerGap(124, Short.MAX_VALUE))
    );

    pack();
}

private void jTextField1ActionPerformed(java.awt.event.ActionEvent evt) {

```

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}

private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {

    int rowCount = tableModel.getRowCount();
    int calculatedCount = 0;

    if (rowCount == 0) {
        javax.swing.JOptionPane.showMessageDialog(this,
            "Сначала добавьте данные через кнопку 'Заполнить'",
            "Информация",
            javax.swing.JOptionPane.INFORMATION_MESSAGE);
        return;
    }

    for (int i = 0; i < rowCount; i++) {

        Object resultValue = tableModel.getValueAt(i, 3);
        boolean hasResult = resultValue != null && !resultValue.toString().trim().isEmpty();

        if (hasResult) {
            continue;
        }

        try {
            Object lowerObj = tableModel.getValueAt(i, 0);
            Object upperObj = tableModel.getValueAt(i, 1);
            Object stepObj = tableModel.getValueAt(i, 2);

            if (lowerObj == null || upperObj == null || stepObj == null ||
                lowerObj.toString().trim().isEmpty() ||
                upperObj.toString().trim().isEmpty() ||
                stepObj.toString().trim().isEmpty()) {
                continue;
            }

            double lowerLimit = Double.parseDouble(lowerObj.toString());
            double upperLimit = Double.parseDouble(upperObj.toString());
            double step = Double.parseDouble(stepObj.toString());

            if (lowerLimit >= upperLimit) {
                tableModel.setValueAt("Ошибка: нижний >= верхний", i, 3);
                continue;
            }

            if (step <= 0) {
                tableModel.setValueAt("Ошибка: шаг <= 0", i, 3);
                continue;
            }

            // вычисл интеграла

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        double result = calculateIntegral(lowerLimit, upperLimit, step);

        tableModel.setValueAt(String.format("%.6f", result), i, 3);
        calculatedCount++;

    } catch (NumberFormatException ex) {
        tableModel.setValueAt("Ошибка формата", i, 3);
    } catch (Exception ex) {
        tableModel.setValueAt("Ошибка", i, 3);
    }
}

if (calculatedCount == 0) {
    javax.swing.JOptionPane.showMessageDialog(this,
        "Нет строк для расчета (все строки уже содержат результаты или имеют ошибки)",
        "Информация",
        javax.swing.JOptionPane.INFORMATION_MESSAGE);
}
}

private void jTable1ComponentHidden(java.awt.event.ComponentEvent evt) {
    // TODO add your handling code here:
}

private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {

    try {
        double lowerLimit = Double.parseDouble(jTextField2.getText().trim());
        double upperLimit = Double.parseDouble(jTextField1.getText().trim());
        double step = Double.parseDouble(jTextField3.getText().trim());

        if (lowerLimit >= upperLimit) {
            javax.swing.JOptionPane.showMessageDialog(this,
                "Нижний предел должен быть меньше верхнего!",
                "Ошибка",
                javax.swing.JOptionPane.ERROR_MESSAGE);
            return;
        }

        if (step <= 0) {
            javax.swing.JOptionPane.showMessageDialog(this,
                "Шаг должен быть положительным числом!",
                "Ошибка",
                javax.swing.JOptionPane.ERROR_MESSAGE);
            return;
        }

        tableModel.addRow(new Object[]{lowerLimit, upperLimit, step, ""});

        jTextField1.setText("");
        jTextField2.setText("");
    }
}

```

```

        jTextField3.setText("");

    } catch (NumberFormatException ex) {
        javax.swing.JOptionPane.showMessageDialog(this,
            "Пожалуйста, введите корректные числовые значения!",
            "Ошибка ввода",
            javax.swing.JOptionPane.ERROR_MESSAGE);
    }
}

private void jButton3ActionPerformed(java.awt.event.ActionEvent evt) {

    int selectedRow = jTable1.getSelectedRow();

    if (selectedRow == -1) {
        javax.swing.JOptionPane.showMessageDialog(this,
            "Пожалуйста, выберите строку для удаления!",
            "Предупреждение",
            javax.swing.JOptionPane.WARNING_MESSAGE);
        return;
    }

    tableModel.removeRow(selectedRow);
}

public double calculateIntegral(double a, double b, double h) {
    double sum = 0.0;
    double x = a;
    int n = (int)((b - a) / h);
    double x0 = 0;
    double x1 = 0;
    for (int i = 0; i < n; i++){
        x0 = a + i * h;
        x1 = a + (i + 1) * h;
        sum += (Math.exp(x0) / x0 + Math.exp(x1) / x1) * h / 2;
    }
    if(x1 < b) sum += (Math.exp(x1) / x1 + Math.exp(b) / b) * h / 2;
    return sum;
}

/**
 * @param args the command line arguments
 */
public static void main(String args[]) {

    try {
        for (javax.swing.UIManager.LookAndFeelInfo info :
            javax.swing.UIManager.getInstalledLookAndFeels()) {
            if ("Nimbus".equals(info.getName())) {
                javax.swing.UIManager.setLookAndFeel(info.getClassName());
                break;
            }
        }
    }
}

```



```
    } catch (ReflectiveOperationException | javax.swing.UnsupportedLookAndFeelException
ex) {
        logger.log(java.util.logging.Level.SEVERE, null, ex);
    }

    java.awt.EventQueue.invokeLater(() -> new l1frame().setVisible(true));
}
```

```
private javax.swing.JButton jButton1;
private javax.swing.JButton jButton2;
private javax.swing.JButton jButton3;
private javax.swing.JLabel jLabel1;
private javax.swing.JLabel jLabel2;
private javax.swing.JLabel jLabel3;
private javax.swing.JLabel jLabel4;
private javax.swing.JLabel jLabel5;
private javax.swing.JLabel jLabel6;
private javax.swing.JScrollPane jScrollPane1;
private javax.swing.JTable jTable1;
private javax.swing.JTextField jTextField1;
private javax.swing.JTextField jTextField2;
private javax.swing.JTextField jTextField3;

}
```